

Little Sub and Sequence

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Codes

AI Code Tips

Description

Little Sub has an integer sequence A .

He will use some bitwise operations like Or, And and Xor(Exclusive Or) to change the sequence. Little Sub wants you to calculate the K th minimum integer in $A[L]..A[R]$ after some operations.

Input

In the first line, it contains two integers n, q ($1 \leq n, q \leq 50000$), indicating the number of sequence integers and the number of operations.

In the second line, it contains n integers $A[i]$, indicating the initial sequence.

The following q lines represents all operations. Each type of operations follows specific formats:

Or X : Change all $A[i]$ to $A[i]$ Or X .

And X : Change all $A[i]$ to $A[i]$ And X .

Xor X : Change all $A[i]$ to $A[i]$ Xor X .

Ask L R K : Ask the K th minimum integer in $A[L]..A[R]$. ($1 \leq K \leq R - L + 1$)

It is guaranteed that $0 \leq A[i], X \leq 2^{31} - 1$.

Output

For each query, output the answer in one line.

Samples

input

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```
10 4
1 2 3 4 5 6 7 8 9 0
Xor 1
And 254
Ask 1 2 2
Ask 1 8 3
```

output

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2
2

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[HZNUOJ](#) is based on [HUSTOJ](#)

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